Recall : Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix.
If a nonzero vector $\mathbf{x}$ in $\mathbb{R}^n$ satisfies $A\mathbf{x} = \lambda\mathbf{x}$ for a scalar λ, then :

- The scalar λ is called an eigenvalue of A.
- The vector $\mathbf{x}$ is called an eigenvector of A with eigenvalue λ.

Example. $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$ with eigenvalue $\lambda = 2$:

$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix} = 2\begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

Eigenspaces

Let A be an $n \times n$ matrix.

If λ is an eigenvalue of A, the set of vectors $\mathbf{x}$ in $\mathbb{R}^n$ satisfying $A\mathbf{x} = \lambda\mathbf{x}$ forms a subspace of $\mathbb{R}^n$. It is called the eigenspace of A corresponding to λ.

- The eigenspace corresponding to λ is identical to $\mathrm{Nul}(A - \lambda\mathrm{I})$.

$$A\mathbf{x} = \lambda\mathbf{x} \Leftrightarrow (A - \lambda\mathrm{I})\mathbf{x} = \mathbf{0}$$

Finding Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix.

- To find the eigenvalues of A, we solve the equation

$$\det(A - \lambda \mathrm{I}) = 0$$

  for λ.

- For each eigenvalue λ, we find a basis for $\mathrm{Nul}(A - \lambda \mathrm{I})$.

The characteristic equation

Let A be an $n \times n$ matrix.

The equation $\det(A - \lambda \mathrm{I}) = 0$ is called the characteristic equation of A.

- The characteristic equation has the following form :

$$\pm\lambda^n + c_{n-1}\lambda^{n-1} + c_{n-2}\lambda^{n-2} + \cdots + c_1\lambda + c_0 = 0$$

Example

Find the eigenvalues of $A = \begin{bmatrix} -4 & 0 & 1 \\ 3 & 2 & -5 \\ -6 & 0 & 1 \end{bmatrix}$.

Solution. First we find the characteristic equation :

$$\begin{aligned}
\det(A - \lambda \mathrm{I}) &= \begin{vmatrix} -4-\lambda & 0 & 1 \\ 3 & 2-\lambda & -5 \\ -6 & 0 & 1-\lambda \end{vmatrix} \\
&= -0\begin{vmatrix} 3 & -5 \\ -6 & 1-\lambda \end{vmatrix} + (2-\lambda)\begin{vmatrix} -4-\lambda & 1 \\ -6 & 1-\lambda \end{vmatrix} - 0\begin{vmatrix} -4-\lambda & 1 \\ 3 & -5 \end{vmatrix} \\
&= (2-\lambda)\Big((-4-\lambda)(1-\lambda) - (1)(-6)\Big) \\
&= (2-\lambda)(\lambda^2 + 3\lambda + 2) = (2-\lambda)(\lambda+1)(\lambda+2)
\end{aligned}$$

Because

$$\lambda^2 + 3\lambda + 2 = 0 \;\Rightarrow\; \lambda = \frac{-3 \pm \sqrt{9-8}}{2} = \begin{cases} -1 \\ -2 \end{cases}$$

Therefore the eigenvalues of A are $2, -2, -1$.

Multiplicity of an eigenvalue

Sometimes the characteristic equation may have repeated roots.

- The number of times an eigenvalue λ appears as a root of the characteristic equation is called its multiplicity.
- If we take multiplicities into account, the characteristic equation of an $n \times n$ matrix has n roots.

Example

Suppose A is a 5×5 matrix whose characteristic equation is $-\lambda^5 - 2\lambda^4 - \lambda^3 = 0$. Find the eigenvalues of A and their multiplicities.

Solution.

$$-\lambda^5 - 2\lambda^4 - \lambda^3 = -\lambda^3(\lambda^2 + 2\lambda + 1) = -\lambda^3(\lambda + 1)^2$$

Therefore the eigenvalues are :

- 0 with multiplicity 3
- -1 with mutiplicity 2

You can see that the total sum of multiplicities is 3+2=5.

Multiplicity and dimension of eigenspace

Theorem

Let A be a square matrix an λ an eigenvalue of A.

- In general we have :

$$\begin{array}{c}\text{Dimension of the eigenspace} \\ \text{corresponding to } \lambda\end{array} \leqslant \text{Multiplicity of } \lambda$$

- The inequality can be strict.

Example. Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. Then :

$$\det(A - \lambda \mathrm{I}) = \begin{vmatrix} 1-\lambda & 0 \\ 1 & 1-\lambda \end{vmatrix} = (1-\lambda)^2$$

Therefore $\lambda = 1$ is an eigenvalue with multiplicity 2.

However, $A\mathbf{x} = \mathbf{x} \Rightarrow (A - \mathrm{I})\mathbf{x} = 0 \Rightarrow \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$x_1 = 0 \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Which means that the eigenspace corresponding to $\lambda = 1$ is one-dimensional.

Example

Let $A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 0 \\ -5 & -5 & -4 \end{bmatrix}$. Calculate the eigenvalues of A and their multiplicities.

$$A - \lambda \mathrm{I} = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 0 \\ -5 & -5 & -4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & 3 & 1 \\ 0 & 1-\lambda & 0 \\ -5 & -5 & -4-\lambda \end{bmatrix}.$$

We use the cofactor expansion across the second row :

$$\begin{vmatrix} 2-\lambda & 3 & 1 \\ 0 & 1-\lambda & 0 \\ -5 & -5 & -4-\lambda \end{vmatrix} = -0 \begin{vmatrix} 3 & 1 \\ -5 & -4-\lambda \end{vmatrix} + (1-\lambda) \begin{vmatrix} 2-\lambda & 1 \\ -5 & -4-\lambda \end{vmatrix} - 0 \begin{vmatrix} 2-\lambda & 3 \\ -5 & -5 \end{vmatrix}$$

$$= (1-\lambda)\Big((2-\lambda)(-4-\lambda) - 1(-5)\Big) = (1-\lambda)(\lambda^2 + 2\lambda - 3)$$

For the second factor we have : $\lambda = \frac{-2 \pm \sqrt{2^2 - (4)(1)(-3)}}{2} = \begin{cases} 1 \\ -3 \end{cases}$

It follows that $(1-\lambda)(\lambda^2 - 2\lambda - 3) = -(\lambda - 1)(\lambda - 1)(\lambda - (-3))$
And the eigenvalues of A are :

- 1 with multiplicity 2,
- -3 with multiplicity 1.

Recall : diagonal matrices

An $n \times n$ matrix is called a diagonal matrix if every entry outside the diagonal of the matrix is zero.

Example.

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$ is a diagonal matrix.

But $$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$ is NOT a diagonal matrix.

Powers of matrices - the diagonal case

Example. Let $A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$. Calculate A^{20}.

Solution.
It is easily seen that

$$A^2 = A \cdot A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} (-1)^2 & 0 & 0 \\ 0 & 2^2 & 0 \\ 0 & 0 & (\frac{1}{2})^2 \end{bmatrix}$$

Similarly

$$A^3 = A \cdot A \cdot A = \begin{bmatrix} (-1)^3 & 0 & 0 \\ 0 & 2^3 & 0 \\ 0 & 0 & (\frac{1}{2})^3 \end{bmatrix}$$

...

We have :

$$A^{20} = \begin{bmatrix} (-1)^{20} & 0 & 0 \\ 0 & 2^{20} & 0 \\ 0 & 0 & (\frac{1}{2})^{20} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1048576 & 0 \\ 0 & 0 & \frac{1}{1048576} \end{bmatrix}$$

The above example shows that we can easily compute any power of a diagonal matrix.

Powers of matrices - the general case

Example

Let $A = \begin{bmatrix} 5 & 4 \\ -8 & -7 \end{bmatrix}$. Calculate A^5.

Solution.

- Suppose we can find an invertible 2×2 matrix P such that $P^{-1}AP$ is diagonal.

For example, consider $P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \Rightarrow P^{-1} = \frac{1}{-1}\begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$

and

$$P^{-1}AP = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 5 & 4 \\ -8 & -7 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix} = D$$

Then

$$P^{-1}AP = D \Rightarrow A = PDP^{-1}$$

and

$$A^5 = (PDP^{-1})(PDP^{-1})(PDP^{-1})(PDP^{-1})(PDP^{-1}) = PD^5P^{-1}$$

$$A^5 = (PDP^{-1})(PDP^{-1})(PDP^{-1})(PDP^{-1})(PDP^{-1}) = PD^5P^{-1}$$

$$\begin{aligned} A^5 &= \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 1^5 & 0 \\ 0 & (-3)^5 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -243 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -243 \\ -1 & 486 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 245 & 244 \\ -488 & -487 \end{bmatrix} \end{aligned}$$

Calculating A^r

Let A be an $n \times n$ matrix. We want to calculate A^r.

- Suppose we can find an invertible $n \times n$ matrix P and a diagonal matrix D such that $A = PDP^{-1}$. Then:

$$A^r = PD^rP^{-1}$$

But how do we find such a matrix P?

Diagonalizable matrices

An $n \times n$ matrix A is called diagonalizable if $A = PDP^{-1}$ where

- an invertible $n \times n$ matrix P, and
- a diagonal $n \times n$ matrix D.

• If A is diagonalizable, then we can calculate powers of A quickly using the following method :

Calculating A^r

Suppose A is diagonalizable, i.e., $A = PDP^{-1}$ with D diagonal. Then:

$$A^r = PD^rP^{-1}$$

Reason :

$$A^r = \underbrace{A \cdot A \cdots A \cdot A}_{r \text{ times}} = \underbrace{(PDP^{-1})(PDP^{-1}) \cdots (PDP^{-1})}_{r \text{ times}} = P\underbrace{D \cdot D \cdots D}_{r \text{ times}}P^{-1} = PD^rP^{-1}$$

How to find D and P?

Example

Is $A = \begin{bmatrix} -7 & -9 \\ 6 & 8 \end{bmatrix}$ diagonalizable? If it is, use diagonalization to evaluate A^{10}.

Diagonalization Algorithm

To diagonalize an $n \times n$ matrix A:

1. Find the eigenvalues of A.
2. Find a basis for each eigenspace.
3. make a list of *all* of the vectors you obtain in this way :

$$\mathbf{v}_1, \ldots, \mathbf{v}_k$$

- If $k < n$, then A is NOT diagonalizable.
- If $k = n$, then A is diagonalizable : take

$$P = [\ \mathbf{v}_1\ \mathbf{v}_2\ \cdots\ \mathbf{v}_k\ ]$$

and the diagonal entries of D are the eigenvalues of A with multiplicities.

- The case $k > n$ is impossible.

Example

Example

Let $A = \begin{bmatrix} -7 & -9 \\ 6 & 8 \end{bmatrix}$. Is A diagonalizable? If so, use diagonalization to calculate A^{10}.

Solution. Finding the eigenvalues of A :

$$A - \lambda \mathrm{I} = \begin{bmatrix} -7 & -9 \\ 6 & 8 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -7-\lambda & -9 \\ 6 & 8-\lambda \end{bmatrix}$$

$$\begin{vmatrix} -7-\lambda & -9 \\ 6 & 8-\lambda \end{vmatrix} = (-7-\lambda)(8-\lambda) - (-9)6 = \lambda^2 - \lambda - 2 = (\lambda+1)(\lambda-2)$$

$$\lambda = \frac{-(-1) \pm \sqrt{1-(4)(-2)}}{2} = \begin{cases} -1 \\ 2 \end{cases}$$

Example (cont.)

Example

Let $A = \begin{bmatrix} -7 & -9 \\ 6 & 8 \end{bmatrix}$. Is A diagonalizable? If so, use diagonalization to calculate A^{10}.

Solution. Finding a basis for each eigenspace :

$\boxed{\lambda = -1}$

$$A - (-1)\mathrm{I} = \begin{bmatrix} -6 & -9 \\ 6 & 9 \end{bmatrix} \qquad \rightsquigarrow \qquad \begin{bmatrix} -6 & -9 \\ 6 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -6 & -9 & 0 \\ 6 & 9 & 0 \end{array}\right] \xrightarrow{\text{Row reduction}} \left[\begin{array}{cc|c} 1 & \frac{3}{2} & 0 \\ 0 & 0 & 0 \end{array}\right]$$

x_1 : basic; x_2 : free.

$\{x_1 = -\frac{3}{2}x_2$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{3}{2}x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -\frac{3}{2} \\ 1 \end{bmatrix}$$

and a basis for the eigenspace corresponding to $\lambda = -1$ is $\left\{ \begin{bmatrix} -\frac{3}{2} \\ 1 \end{bmatrix} \right\}$

Example (cont.)

Example

Let $A = \begin{bmatrix} -7 & -9 \\ 6 & 8 \end{bmatrix}$. Is A diagonalizable? If so, use diagonalization to calculate A^{10}.

Finding a basis for each eigenspace :

$\boxed{\lambda = 2}$

$$A - 2\mathrm{I} = \begin{bmatrix} -9 & -9 \\ 6 & 6 \end{bmatrix} \qquad \rightsquigarrow \qquad \begin{bmatrix} -9 & -9 \\ 6 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -9 & -9 & 0 \\ 6 & 6 & 0 \end{array}\right] \xrightarrow{\text{Row reduction}} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

x_1 : basic; x_2 : free.

$\{x_1 = -x_2$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

and a basis for the eigenspace corresponding to $\lambda = 2$ is $\left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$

Example (cont.)

Example

Let $A = \begin{bmatrix} -7 & -9 \\ 6 & 8 \end{bmatrix}$. Is A diagonalizable? If so, use diagonalization to calculate A^{10}.

Listing all of the vectors we obtained : $\mathbf{v}_1 = \begin{bmatrix} -\frac{3}{2} \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

A is 2×2 and the list contains two vectors. Therefore A is diagonalizable.

$$P = \begin{bmatrix} -\frac{3}{2} & -1 \\ 1 & 1 \end{bmatrix} \Rightarrow P^{-1} = \frac{1}{-\frac{1}{2}} \begin{bmatrix} 1 & 1 \\ -1 & -\frac{3}{2} \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 2 & 3 \end{bmatrix}$$

We have :

$$P^{-1}AP = \begin{bmatrix} -2 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -7 & -9 \\ 6 & 8 \end{bmatrix} \begin{bmatrix} -\frac{3}{2} & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$$

Example (cont.)

Example

Let $A = \begin{bmatrix} -7 & -9 \\ 6 & 8 \end{bmatrix}$. Is A diagonalizable? If so, use diagonalization to calculate A^{10}.

Finally, we calculate A^{10} :

$$P = \begin{bmatrix} -\frac{3}{2} & -1 \\ 1 & 1 \end{bmatrix} \Rightarrow P^{-1}AP = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow A = P \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix} P^{-1}$$

$$A^{10} = P \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}^{10} P^{-1} = \begin{bmatrix} -\frac{3}{2} & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} (-1)^{10} & 0 \\ 0 & 2^{10} \end{bmatrix} \begin{bmatrix} -2 & -2 \\ 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{3}{2} & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1024 \end{bmatrix} \begin{bmatrix} -2 & -2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} -2045 & -3069 \\ 2046 & 3070 \end{bmatrix}$$

Another example

Example

Let $A = \begin{bmatrix} 3 & -3 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$. Is A diagonalizable?

Solution.

The matrix is triangular and its eigenvalues are 3 and -1. We find a basis for each eigenspaces :

$\boxed{\lambda = -1}$

$$A - (-1)\mathrm{I} = \begin{bmatrix} 3 & -3 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 4 & -3 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 4 \end{bmatrix} \quad \begin{bmatrix} 4 & -3 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 4 & -3 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 4 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & -\frac{3}{4} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{cases} x_1 - \frac{3}{4}x_2 = 0 \\ x_3 = 0 \end{cases}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{3}{4}x_2 \\ x_2 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} \frac{3}{4} \\ 1 \\ 0 \end{bmatrix} \Rightarrow \boxed{\text{Basis for eigenspace : } \left\{ \begin{bmatrix} \frac{3}{4} \\ 1 \\ 0 \end{bmatrix} \right\}.}$$

Another example (cont.)

Example

Let $A = \begin{bmatrix} 3 & -3 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$. Is A diagonalizable?

$\boxed{\lambda = 3}$

$$A - 3\mathrm{I} = \begin{bmatrix} 3 & -3 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 0 & -3 & 1 \\ 0 & -4 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & -3 & 1 \\ 0 & -4 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 0 & -3 & 1 & 0 \\ 0 & -4 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{cases} x_2 = 0 \\ x_3 = 0 \end{cases}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ 0 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \boxed{\text{Basis for eigenspace} : \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}.}$$

- We now list all of the vectors we have obtained : $\mathbf{v}_1 = \begin{bmatrix} \frac{3}{4} \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

We only have 2 vectors, but A is 3×3. Therefore A is NOT diagonalizable.

Theorem

Let A be an $n \times n$ matrix. If A has exactly n distinct eigenvalues, then A is diaonalizable.

Reason : For each eigenvalue, we obtain one eigenvector, and therefore the list of vectors

$$\mathbf{v}_1, .., \mathbf{v}_k$$

that we obtain has exactly n elements, i.e., $k = n$.

Example

Is $A = \begin{bmatrix} 3 & -1 & 0 & 2 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & -1 & 5 \\ 0 & 0 & 0 & 5 \end{bmatrix}$ diagonalizable?

Solution. Yes! In fact since A is triangular, its eigenvalues are 3,2,-1,5.
A is 4×4 and has exactly 4 distinct eigenvalues.
Therefore A is diagonalizable.

Example

Let $A = \begin{bmatrix} -5 & 0 & -3 \\ -6 & 1 & -3 \\ 6 & 0 & 4 \end{bmatrix}$. We know that the eigenvalues of A are 1 and -2. Is A diagonalizable?

Solution. First we find a basis for each eigenspace :

$\boxed{\lambda = 1}$

$$A - \mathrm{I} = \begin{bmatrix} -5 & 0 & -3 \\ -6 & 1 & -3 \\ 6 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -6 & 0 & -3 \\ -6 & 0 & -3 \\ 6 & 0 & 3 \end{bmatrix} \quad \begin{bmatrix} -6 & 0 & -3 \\ -6 & 0 & -3 \\ 6 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -6 & 0 & -3 & 0 \\ -6 & 0 & -3 & 0 \\ 6 & 0 & 3 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} -6 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to -\frac{1}{6} R_1} \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

x_1 : basic; x_2, x_3 : free; $\quad x_1 + \frac{1}{2} x_3 = 0$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix} \Rightarrow \boxed{\text{Basis for the eigenspace is } \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix} \right\}}$$

Example

Let $A = \begin{bmatrix} -5 & 0 & -3 \\ -6 & 1 & -3 \\ 6 & 0 & 4 \end{bmatrix}$. We know that the eigenvalues of A are 1 and -2. Is A diagonalizable?

$\boxed{\lambda = -2}$

$$A - (-2)\mathrm{I} = \begin{bmatrix} -5 & 0 & -3 \\ -6 & 1 & -3 \\ 6 & 0 & 4 \end{bmatrix} - \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} -3 & 0 & -3 \\ -6 & 3 & -3 \\ 6 & 0 & 6 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 0 & -3 \\ -6 & 3 & -3 \\ 6 & 0 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -3 & 0 & -3 & 0 \\ -6 & 3 & -3 & 0 \\ 6 & 0 & 6 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 + 2R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} -3 & 0 & -3 & 0 \\ 0 & 3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow[R_1 \to -\frac{1}{3}R_1]{R_2 \to \frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

x_1, x_2 : basic; x_3 : free; $\begin{cases} x_1 + x_3 = 0 \\ x_2 + x_3 = 0 \end{cases}$ $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_3 \\ -x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \Rightarrow \boxed{\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}}$

Our list of vectors : $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ Therefore A is **diagonalizable.**

Example

Let $A = \begin{bmatrix} -5 & 0 & -3 \\ -6 & 1 & -3 \\ 6 & 0 & 4 \end{bmatrix}$. We know that the eigenvalues of A are 1 and -2. Is A diagonalizable?

Our list of vectors :

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \ \mathbf{v}_2 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}, \ \mathbf{v}_3 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \ \Rightarrow \ P = \begin{bmatrix} 0 & -\frac{1}{2} & -1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}, \ P^{-1} = \begin{bmatrix} -2 & 1 & -1 \\ 2 & 0 & 2 \\ -2 & 0 & -1 \end{bmatrix}$$

and we have

$$P^{-1}AP = \begin{bmatrix} -2 & 1 & -1 \\ 2 & 0 & 2 \\ -2 & 0 & -1 \end{bmatrix} \begin{bmatrix} -5 & 0 & -3 \\ -6 & 1 & -3 \\ 6 & 0 & 4 \end{bmatrix} \begin{bmatrix} 0 & -\frac{1}{2} & -1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Question. What happens if we change the order of $\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$?

Answer. The order of diagonal entries of D changes.

For example, if $P = [\mathbf{v}_2 \ \mathbf{v}_3 \ \mathbf{v}_1] = \begin{bmatrix} -\frac{1}{2} & -1 & 0 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ then $P^{-1} = \begin{bmatrix} 2 & 0 & 2 \\ -2 & 0 & -1 \\ -2 & 1 & -1 \end{bmatrix}$ and

$$P^{-1}AP = \begin{bmatrix} 2 & 0 & 2 \\ -2 & 0 & -1 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} -5 & 0 & -3 \\ -6 & 1 & -3 \\ 6 & 0 & 4 \end{bmatrix} \begin{bmatrix} -\frac{1}{2} & -1 & 0 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Summary and final remarks

Let A be an $n \times n$ matrix.

- Method for calculating A^r : If $A = PDP^{-1}$, then $A^r = PD^rP^{-1}$.
- Having A, how can we find P and D ?

1. Find the eigenvalues of A.
2. Find a basis for each eigenspace.
3. make a list of *all* of the vectors you obtain in this way :

$$\mathbf{v}_1, \ldots, \mathbf{v}_k$$

- If $k < n$, then A is NOT diagonalizable.
- If $k = n$, then A is diagonalizable : We take

$$P = [\ \mathbf{v}_1\ \mathbf{v}_2\ \cdots\ \mathbf{v}_k\ ]$$

and the diagonal entries of D are the eigenvalues of A with multiplicities.

- The case $k > n$ is impossible.